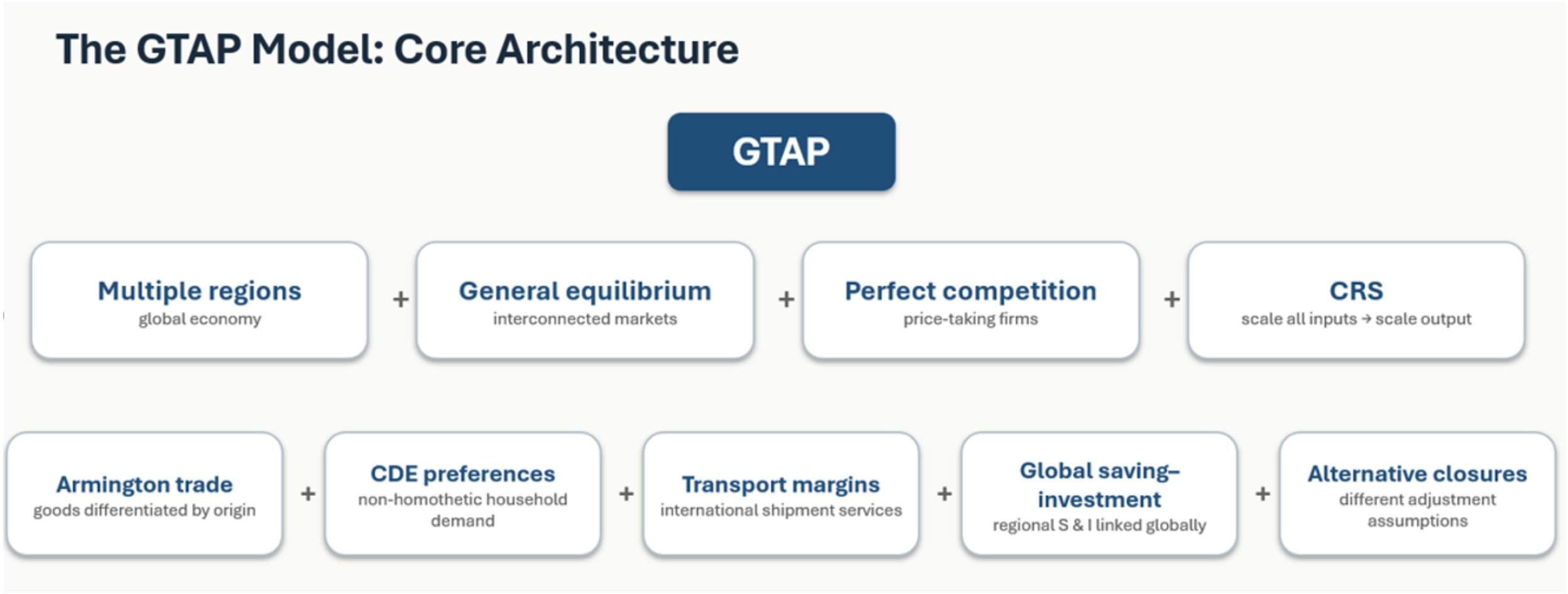
The GTAP Model: Core Architecture



Linearized Static AGE vs Non-Linear Dynamic MIRAGE

What We Need in GTAP: From Model Structure to Simulation

Sets define the economy • Subsets identify special groups • Variables change • Elasticities govern responses

1

SETS

Who / what exists?

Dimensions of the model

REG • COMM • ENDW

e.g., DVCUSA, Oil, Capital

2

SUBSETS

Which members are special?

Selected members of a larger set

FUEL • EGY • CTAX

e.g., Oil & Gas within energy

3

VARIABLES

What can change?

Quantities, prices, trade, policy & technology

qo • qxs • prices • taxes

EXO or ENDO depends on closure

4

ELASTICITIES

How strongly do agents respond?

Behavioral response parameters

ESUBD • ESUBM • ESUBVA

e.g., Oil ESUBD = 5.20

GTAP Simulation: Concepts You Must Get Right

These concepts determine what you can shock, what GTAP solves, and how to interpret the experiment.

DIMENSION

Where does the variable live?

Indices define the component: commodity × region × source/destination.

qxs(Oil, USA, EU) is 3-D

EXOGENOUS

What do I determine?

Fixed outside the model under the active closure; eligible for a direct shock.

Policy, technology, population

Quick Quiz

ENDOGENOUS

What does GTAP determine?

Solved jointly by the equilibrium system. Leave it endogenous when it is an outcome.

Output, prices, trade, welfare

Quiz scores

SHOCK

What changes from the baseline?

A specified change to an exogenous variable/component.

tm = –10%, productivity = +5%

SWAP

When do I reverse roles?

Make an endogenous target exogenous and free a suitable exogenous instrument.

Target qxs = +5% → instrument becomes ENDO

CLOSURE

Who is EXO vs ENDO?

The complete assignment of variable components that defines the economic experiment.

Closure tells us what is allowed to adjust after the shock.

VARIABLE + DIMENSION → CLOSURE → EXO? SHOCK DIRECTLY | ENDO? OBSERVE IT | ENDO TARGET? SWAP → SOLVE

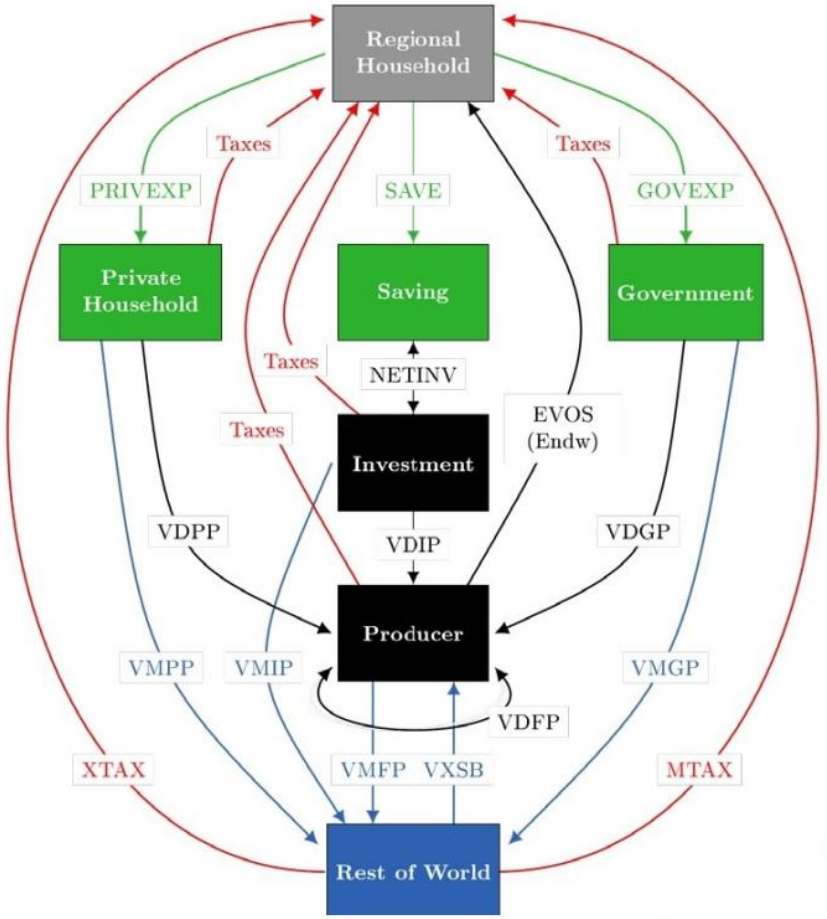
How does energy demand disturbances reshape trade dynamics, efficiency metrics, and economic sustainability in varying regional contexts?

- **Key Takeaway**

  From energy shocks to economic shifts, how to map these ripple effects.

DP: Domestic purchase, PP: Producer Price

Category	Description
VDPP	DP by households at PP
VDFP	DP by Firms at PP
VDGP	DP by Government at PP
VDIP	DP by Investment at PP
ESBD	Armington CES Domestic Allocation
ESBM	Armington CES Reg. Imports Allocation
ESBT	CES b/w Primary & Intermediate Inputs
ESBV	CES b/w Primary Factors in Production
ETRE	CET b/w Sectors Sluggish Adjustments



Standard GTAP Model

- Data Source: GTAP-E TP16 Version is used to analyze this study.
- Regions:

  a) Developed: USA, EU, JPN, EEFSU, ROA1

  b) Developing: CHIND, EEX, RoW
- Sectors: There are 8 sectors in GTAP-E TP 16, however, this study is focused on Agriculture, Electricity, Coal, Oil, Gas, and En_Int_Ind, etc.
- Shocks: Six shocks are implemented.

  a) Oil Demand Disruptions, e.g. Shock qdem("oil","DVCUSA") = -15;

b) Import Inefficiency ams("En_Int_ind","DVCUSA","DPCCHIND") = -5;

c) Oil Demand Disruptions, e.g. Shock qdem("oil","DPCHIIND") = -6;

- Swap: In this study, five swaps are considered.

a) e.g. Swap qdem("oil","DVCUSA") = aoall("oil","DVCUSA");

Table 1: Energy Demand Declines by Region in 2020 (% Change from 2019)
Parentheses shows amount of shock in percentage

Region	Oil & Petroleum	Coal	Natural Gas	Electricity	Sources
USA	↓13% (15%)	↓19% (20%)	↓2% (5%)	↓4% (5%)	EIA
EU	↓12.9% (15%)	↓18–20% (20%)	↓2.6% (5%)	↓12% (15%)	EEC
Japan	↓7.1% (10%)	↓14.9%(15%)	↓8.3% (10%)	↓2.1% (5%)	METI
EEFSU	↓9% (10%)	↓10% (15%)	↓7% (10%)	↓5% (7%)	IEA
China	↓9% (30%)	↓8% (10%)	↓2% (5%)	↓11% (30%)	Hoang et al. (2021)
India	↓30% (30%)	↓10% (10%)	↓2% (5%)	↓30% (30%)	Hoang et al. (2021)
Energy Exporters (EEx)	↓9% (10%)	↓10% (10%)	↓7% (10%)	↓5% (7%)	IEA
Rest of World (RoW)	↓9% (10%)	↓10% (10%)	↓7% (10%)	↓5% (7%)	IEA

Scenario 1: Disruptions in US Energy Sectors

Scenario 2: Impose Tariff Through Efficiency Shock (ams)

Scenario 3: Demand Disturbance in Developing Countries (CHIIND)

- **Scenario 1: Disruptions in US Energy Sectors**

USA	Oil	Coal	Oil Pcts	Electricity	Gas	AGR
qpd	-43.5	-37.6	-16.7	-7.47	-6.07	-0.064

Table 2: US Private Household Demand (qpd(i,s))

Sector	AGR	Coal	Oil	Gas	EnInt	Otind	OilPct	Elec.	Total
Sector	0	-36.40	-31.41	-9.00	0	0	-5.11	-0.87	-82.79

Table 3: Technical Efficiency, aoall

AGR	Coal	Oil	Gas	EnInt	Othind	OilPct	Elec.
0.554	-33.546	-46.288	-21.661	0.142	-0.071	-19.263	-5.227

Table 4: Sector-wise Production Performance

Sector	DVCRoA	DPCROW	DPCCHIND	DVCJPN	DPCEEx	DVCUSA	DVCEU	DVCROW	DVCFSU
AGR	-2.19	-2.09	-1.00	-1.56	-1.02	1.68	-1.47	-1.98	-1.68
Coal	198.4	199.9	213.5	209.58	202.9	-79.7	203.6	203.2	208.0
Oil	9.3	30.9	40.0	41.2	23.6	-74.3	35.7	30.6	40.2
Gas	26.8	60.7	67.7	42.5	35.0	-60.2	68.9	58.9	69.2
EnInd	-1.8	-2.7	-0.4	-1.3	-1.4	1.6	-0.7	-2.8	-0.7
OlPct	9.5	12.0	14.6	11.03	10.0	-33.6	10.4	12.2	11.9
Elec	12.9	11.7	15.8	12.8	11.0	-20.0	14.8	12.6	14.9

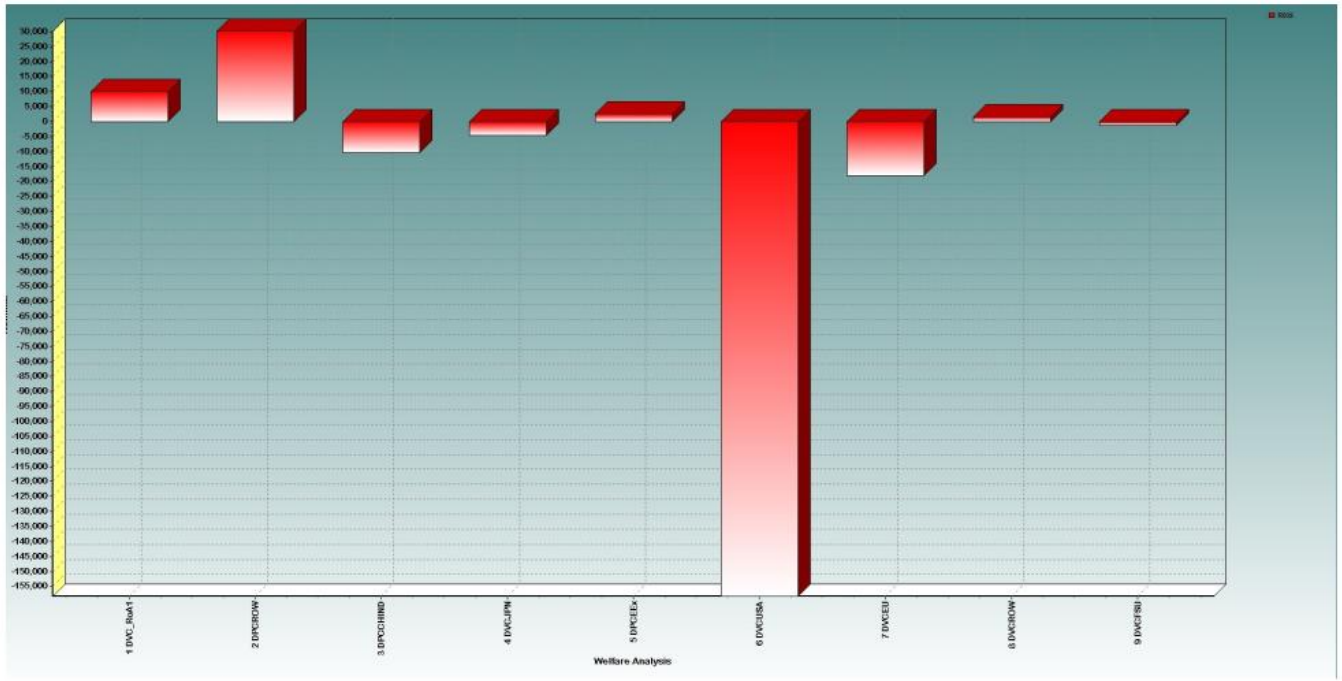
Table 5: Export Trends Amidst Production Shocks (qxs(i,r,s))

Sector	AGR	Coal	Oil	Gas	EnInt	OthIndr	OilPct	Elec.
qim	-1.59	202.01	22.04	28.94	-1.5	-3.6	11.6	12.9
e1_SHRIFM	-0.765	201.977	22.045	28.798	-1.197	-2.305	9.305	9.993
e1_SHRIPM	-0.826	0.037	0.000	0.145	-0.290	-1.277	2.252	2.924
e1_SHRIGM	0.000	0	0	0	-0.000	-0.031	0	0

Table 6: Aggregate Import Demand (qim)

Category	Value
DVCRoA	9989.3
DPCROW	30213.5
DPCCHIND	-10293.9
DVCJPN	-4449.7
DPCEEx	2527.0
DVCUSA	-158261.2
DVCEU	-17981.6
DVCROW	1751.0
DVCFSU	-1079.1

Table 10: Welfare Comparison



Welfare Trend